

Ryan Cho

Principal Software Engineer · Full-Stack Architecture & AI-Enabled Platforms

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PROFESSIONAL SUMMARY

Software engineer with nearly a decade of experience architecting cloud-native, API-first platforms in regulated enterprise environments. Leads full-stack delivery across React, TypeScript, Java, and Python on AWS and Azure, integrating generative AI into production workflows. Drives distributed system design, microservices modernization, and CI/CD automation while mentoring engineers and aligning technical roadmaps with product strategy.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Defined **API-first microservices architecture** for portfolio reporting and financial analytics platforms serving 200+ internal users, establishing service boundaries, contract standards, and design review gates adopted across three engineering squads.
- Built **React and TypeScript** dashboards and admin tooling with role-based access controls, reducing manual reconciliation workflows by 40% through real-time data visualization and configurable reporting views.
- Developed **Java and Python backend services** exposing REST and GraphQL endpoints for portfolio data, trade lifecycle events, and compliance reporting, achieving sub-200ms p95 latency under peak batch and interactive load.
- Integrated **generative AI capabilities** into analyst workflows using LLM APIs and prompt orchestration, enabling natural-language query of financial datasets with audit logging and human-in-the-loop validation for regulated outputs.
- Led **cloud-native migration** of legacy monolith components to containerized services on AWS and Azure, implementing Kubernetes deployments, Terraform infrastructure, and blue-green release pipelines with zero-downtime cutovers.
- Championed **CI/CD and observability** standards including automated test gates, Datadog dashboards, and distributed tracing, cutting production incident mean time to resolution by 35% across core trading and reporting services.
- Designed **event-driven integrations** with Kafka and REST webhooks connecting custodian feeds, market data vendors, and internal ledger systems, ensuring idempotent processing and fault-tolerant retry semantics at enterprise scale.
- Mentored **six engineers** across frontend, backend, and platform workstreams, conducting architecture reviews, pair programming sessions, and sprint planning that improved on-time epic delivery from 72% to 91% over four quarters.
- Partnered with product owners and compliance stakeholders to translate **business-critical requirements** into technical roadmaps, balancing scalability, security, and delivery speed for multi-tenant SaaS features in a highly regulated environment.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Contributed to **distributed trading and risk platforms** supporting fixed-income and equities desks, implementing Java microservices that processed high-volume market events with strict latency and auditability requirements.
- Developed **REST and GraphQL APIs** for position, exposure, and reference data services consumed by internal portals and downstream analytics pipelines across multiple business units.
- Built **Angular and React** interfaces for operations and control teams, replacing spreadsheet-driven workflows with validated web forms, approval chains, and real-time status tracking for trade lifecycle management.
- Implemented **CI/CD pipelines** with Jenkins and GitLab, automating build, security scan, and deployment stages that reduced release cycle time from two weeks to three days for tier-two application teams.

- Supported **cloud modernization initiatives** on AWS, containerizing stateless services and establishing infrastructure-as-code patterns that improved environment parity between development, UAT, and production.
- Designed **SQL and NoSQL data models** for trade capture and reporting stores, optimizing query performance and indexing strategies that lowered batch reconciliation runtime by 25% on nightly jobs.
- Collaborated in **Agile Scrum delivery**, participating in sprint planning, backlog refinement, and cross-functional design reviews with architects, QA, and business analysts on enterprise solution releases.
- Enhanced **monitoring and alerting** with Prometheus and Grafana dashboards tied to on-call runbooks, improving early detection of integration failures across vendor and internal messaging interfaces.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Developed **React components** for a restaurant inventory management web application, implementing responsive layouts and client-side validation that improved data entry accuracy for floor staff.
- Built **Node.js REST endpoints** connecting the frontend to PostgreSQL-backed inventory and order data, following API contract conventions established by the senior engineering team.
- Participated in **code reviews and Agile ceremonies**, delivering sprint commitments on schedule while incorporating feedback on test coverage, error handling, and accessibility improvements.
- Wrote **unit and integration tests** with Jest and Supertest, increasing coverage on core service modules and reducing regression defects found during QA cycles.
- Assisted with **CI pipeline configuration** for automated builds and deployments to staging environments, documenting setup steps for future intern onboarding.
- Refactored legacy **JavaScript modules** into typed TypeScript utilities, improving maintainability and reducing runtime errors in shared authentication and session handling logic.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive marketing pages** for a nonprofit research foundation using HTML, CSS, and JavaScript, ensuring cross-browser compatibility and mobile-first layout standards.
- Integrated **CMS content workflows** allowing non-technical staff to publish articles and event listings without developer intervention for routine updates.
- Implemented **contact and newsletter forms** with client-side validation and server-side submission handling, improving lead capture reliability for outreach campaigns.
- Optimized **page load performance** through image compression, lazy loading, and minified assets, reducing average load time on key landing pages.
- Collaborated with **design stakeholders** to translate wireframes into accessible web components aligned with the organization's brand guidelines.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, Angular, JavaScript, HTML5, CSS3, Responsive Web Design, Component Architecture

Backend: Java, Python, Node.js, REST APIs, GraphQL, gRPC, Microservices, Event-Driven Architecture, Kafka

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL Query Optimization, NoSQL Data Modeling

Cloud & DevOps: AWS, Azure, Google Cloud Platform, Kubernetes, Docker, Terraform, CI/CD, GitLab, Jenkins

Monitoring & Tools: Datadog, Prometheus, Grafana, Distributed Tracing, OpenAI API, LLM Integration, Git, Cursor, Copilot

Workflow & PM: Agile, Scrum, Sprint Planning, Design Reviews, Technical Roadmapping, Cross-Functional Collaboration, Mentorship